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Deep learning framework for rapid aberration correction in reflection matrix microscopy

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Abstract: Aberrations caused by angle-dependent phase distortions in scattering media degrade image quality in optical imaging. In this study, we present a deep learning-based aberration correction method for reflection matrix microscopy that iteratively corrects both input and output aberrations. By leveraging the structural consistency of output aberrations across different incident wavevectors, we train a U-Net-based model to predict and correct aberrations directly from the reflection matrix. Our iterative inference and correction process effectively eliminates round-trip aberrations, significantly enhancing imaging quality. To further improve efficiency, we introduce a covariance matrix-based training strategy, eliminating the need for explicit input aberration correction and reducing iteration time by half. Our approach achieves a 100-fold computational speedup over conventional wave correlation-based algorithms while maintaining high correction accuracy. We validate our method through numerical simulations and experimental data, demonstrating robustness across various aberration conditions. This deep-learning framework enables real-time, label-free imaging, overcoming computational bottlenecks in aberration correction and paving the way for rapid, high-resolution imaging in biomedical applications where real-time aberration correction is essential.

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1. Introduction

Waves propagating through scattering media experience angle-dependent phase retardations, termed aberrations, which distorts wavefront and impairs image quality. Accurately identifying and correcting aberrations is essential for maximizing the performance of optical imaging techniques. Various aberration correction strategies have been developed to address these challenges [1]. A widely used approach involves iteratively optimizing the wavefront using image intensity or sharpness as an indirect measure of aberrations [2–7]. Another method employs wavefront sensing techniques, such as Shack–Hartmann sensors [8] or coherence-gated approaches [9,10], to directly measure aberrations from backscattered waves [8–11]. These techniques are particularly effective in fluorescence imaging, where excitation and emission can be easily distinguished due to their differing wavelengths, and where correcting either input or output aberrations is often sufficient. However, in label-free reflectance imaging, both input and output aberrations contribute to image quality while making their separation is more complex due to their same wavelength.

Reflection matrix microscopy has emerged as a powerful technique for aberration correction in reflectance imaging. In particular, closed-loop accumulation of single scattering (CLASS)-based methods leverage the reflection matrix to precisely correct aberrations, significantly improving deep-tissue imaging quality [12]. Despite their accuracy, CLASS algorithms are computationally intensive and time-consuming, limiting their applicability in real-time imaging.

Recent advancements in deep learning have revolutionized various fields, including optical imaging. Building on this progress, deep-learning models have been employed to estimate

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aberrations using diverse approaches. Some methods rely on point spread functions (PSFs) obtained from known reference structures, such as fluorescent beads, sparsely distributed emitters, or artificial guide stars, to retrieve pupil phase aberrations [13–19], while others predict aberrations based on image quality metrics [20,21]. Additionally, deconvolution-like techniques have been developed to extract PSFs directly from images without requiring guide stars [22,23]. Compared to conventional wavefront sensing and correction methods, these deep learning-based approaches provide significant advantages in computational efficiency and processing time. While such strategies have been widely explored in fluorescence imaging, their application to reflection-based imaging remains largely unexplored.

In this paper, we present a deep-learning-based aberration correction approach for reflection matrix microscopy. A key physical property of the reflection matrix is that the output aberration remains consistent across reflected waves originating from different incident wavevectors. Leveraging this characteristic, we train a U-Net-based model to predict the output aberration directly from the reflection matrix. To correct both input and output aberrations, we apply the trained U-Net iteratively to the reflection matrix and its transpose. Additionally, we extend the correction process to the covariance matrix, which inherently removes the need for explicit input aberration correction, thereby reducing process time by half.

A major contribution of this study is the significant improvement in computational efficiency. Once trained, our deep-learning model performs aberration correction orders of magnitude faster than traditional wave correlation-based methods, achieving a 100-fold speed increase over CLASS algorithms. This breakthrough enables real-time aberration correction, making *in vivo* and *in situ* imaging more feasible. By integrating the precision of reflection matrix analysis with the speed of deep learning, our approach paves the way for real-time, label-free imaging in biomedical research and diagnostics.

2. Deep learning approach

When a plane wave with a specific transverse wave vector $\mathbf{k}_\mathrm{i}$ is incident on a target object embedded within a scattering medium, it propagates through the medium while experiencing phase retardation $\phi_\mathrm{i}(\mathbf{k}_\mathrm{i})$, induced by angle-dependent inhomogeneities. Although the incident wave scatters into multiple wavevector components, we focus on the ballistic (or zeroth-order) component—i.e., the wave that retains its original wavevector $\mathbf{k}_\mathrm{i}$—while treating all other multiply scattered components as noise. After reaching the object, the ballistic wave is reflected and redistributed into various output wave vectors $\mathbf{k}_\mathrm{o}$. When the object is sufficiently thin or depth-sectioned (e.g., via time-gating or confocal gating), it can be modeled as a 2D amplitude reflectance function $O(x, y)$. Then, the amplitude of reflected wave with wave vector $\mathbf{k}_\mathrm{o}$ is given by $\tilde{O}(\mathbf{k}_\mathrm{o} - \mathbf{k}_\mathrm{i})$, where $\tilde{O}$ is the 2D Fourier transform of O. On the return path, the ballistic component of each output wave vector component $\mathbf{k}_\mathrm{o}$ gains an additional phase shift $\phi_\mathrm{o}(\mathbf{k}_\mathrm{o})$, also caused by the scattering medium. As a result, the spatial frequency spectrum of the reflected wave is expressed as:

$$\tilde{E}(\mathbf{k}_\mathrm{o}; \mathbf{k}_\mathrm{i}) = e^{i\phi_\mathrm{o}(\mathbf{k}_\mathrm{o})}\tilde{O}(\mathbf{k}_\mathrm{o} - \mathbf{k}_\mathrm{i})e^{i\phi_\mathrm{i}(\mathbf{k}_\mathrm{i})}. \tag{1}$$

To retrieve the target object from the measured reflected wave $\tilde{E}(\mathbf{k}_\mathrm{o}; \mathbf{k}_\mathrm{i})$, it is essential to correct the output aberration, as it distorts the object image. Since the input aberration only introduces an overall phase shift, it does not affect image formation in individual reflected waves. However, its presence hinders the effectiveness of the synthesis of multiple images, known as Collective Accumulation of Single Scattering (CASS) microscopy [24], a critical method for reducing multiple scattering noise induced by the scattering medium and enhancing spatial frequency bandwidth.

In this study, we utilize a reflection matrix $\boldsymbol{R}$, whose elements are composed of $\tilde{E}(\mathbf{k}_\mathrm{o}; \mathbf{k}_\mathrm{i})$ taken for a set of $\mathbf{k}_\mathrm{i}$, to identify and correct aberrations. The reflection matrix can be represented as a

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2-dimensional matrix, whose column and row indices correspond to $\mathbf{k}_\mathrm{i}$ and $\mathbf{k}_\mathrm{o}$:

$$\boldsymbol{R} = \boldsymbol{P}_\mathrm{o}\boldsymbol{O}\boldsymbol{P}_\mathrm{i}^T. \tag{2}$$

Here $\boldsymbol{P}_\mathrm{o}$ and $\boldsymbol{P}_\mathrm{i}^T$ are diagonal matrices with diagonal elements $e^{i\phi_\mathrm{o}(\mathbf{k}_\mathrm{o})}$ and $e^{i\phi_\mathrm{i}(\mathbf{k}_\mathrm{i})}$. $\boldsymbol{O}$ is a Toeplitz matrix, characterized by constant values along its diagonals, with its elements determined by the object spectrum $\tilde{O}(\mathbf{k}_\mathrm{o} - \mathbf{k}_\mathrm{i})$.

In Eq. (1), it is noteworthy that the reflected waves share a common output aberration $\phi_\mathrm{o}(\mathbf{k}_\mathrm{o})$, regardless of the incident wavevectors $\mathbf{k}_\mathrm{i}$. This is illustrated in Fig. 1(b), which shows that while $\phi_\mathrm{o}(\mathbf{k}_\mathrm{o})$ remains unchanged, $\phi_\mathrm{i}(\mathbf{k}_\mathrm{i})$ and $\tilde{O}(\mathbf{k}_\mathrm{o} - \mathbf{k}_\mathrm{i})$ vary for three different values of $\mathbf{k}_\mathrm{i}$. Leveraging this characteristic, we employed U-Net-based deep learning model [25], which is well-suited for image-to-image regression tasks due to its encoder–decoder structure with skip connections that preserve spatial detail while enabling deep feature extraction. We trained the model to identify the common aberration factor $e^{i\phi_\mathrm{o}(\mathbf{k}_\mathrm{o})}$, which is multiplicatively consistent across reflected waves for all incident wave vectors, from the recorded reflection matrix.

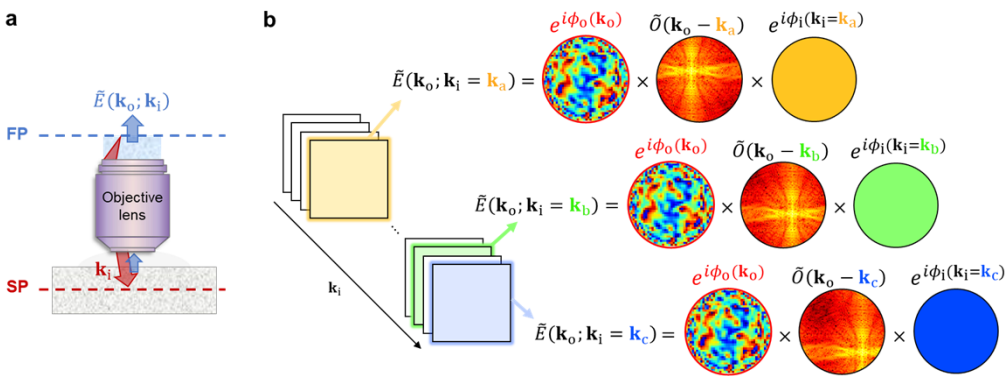

Fig. 1. Aberrations in the reflection matrix. (a) Schematic representation of the setup for measuring the reflection matrix. (b) Visualization of the reflected wave $\tilde{E}(\mathbf{k}_\mathrm{o};\mathbf{k}_\mathrm{i})$ for representative incident wave vectors $\mathbf{k}_\mathrm{a}$, $\mathbf{k}_\mathrm{b}$, and $\mathbf{k}_\mathrm{c}$. Each reflected wave exhibits the same output aberration ϕ_o, while the input aberration ϕ_i varies across the different incident wave vectors, altering the overall phase of each reflected wave. The contribution of the object spectrum $\tilde{O}$ also varies as it shifts relative to the incident wavevector.

To train the network, we prepared two components: the model input and the ground truth. The model input is a reflection matrix, which is computationally generated according to Eq. (1) with parameters including a wavelength of 800 nm, a field of view of 40 µm, and a numerical aperture of 0.4. The 2D spatial frequency space is represented on a 40×40 pixel grid. Both input and output aberrations (ϕ_i and ϕ_o) are generated using a random linear combination of Zernike polynomials. The maximum Zernike order is randomly selected between 10 and 60, with coefficients for each order chosen randomly. The MNIST dataset is used as the object function $O(x, y)$. The corresponding output aberration factor $e^{i\phi_\mathrm{o}}$ serves as the ground truth. Since both the model input and the ground truth are complex numbers, they are decomposed into real and imaginary components, as the PyTorch framework used in this study is optimized for real numbers.

The U-Net architecture used in our model features a contracting path with three stages, where the number of filters increases sequentially from 64 to 128 and then to 256. The bottleneck layer, with 512 filters, connects the contracting and expanding paths. In the expanding path, the number of filters decreases symmetrically from 256 to 128 and finally to 64. The model takes a reflection matrix of size $1600 \times 40 \times 40$ pixels as input and predicts the corresponding aberration map of

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size 40×40 pixels:

$$e^{i\hat{\phi}_o(\mathbf{k}_o)} = UNet(\boldsymbol{R}). \tag{3}$$

Training is conducted by minimizing a loss function L, which is defined as the Pearson correlation between the predicted aberration factor $e^{i\hat{\phi}_o(\mathbf{k}_o)}$ and the ground truth $e^{i\phi_o(\mathbf{k}_o)}$:

$$L = -\left|\frac{\mathbb{E}[(e^{i\phi_o} - \mu(e^{i\phi_o}))(e^{i\hat{\phi}_o} - \mu(e^{i\hat{\phi}_o}))^*]}{\sigma(e^{i\phi_o})\sigma(e^{i\hat{\phi}_o})}\right|, \tag{4}$$

where $\mathbb{E}[\cdot]$ denotes the expectation, $\mu(\cdot)$ represents the mean, and $\sigma(\cdot)$ is the standard deviation.

3. Finding the output aberrations in the absence of input aberrations

To illustrate the methodology for identifying the common factor of output aberration $e^{i\phi_o}$, we initially trained the model using computationally generated reflection matrices containing only output aberrations. The training dataset consisted of 1000 samples, while the test dataset included 100 datasets. The resulting loss curve is shown in Fig. 2(f).

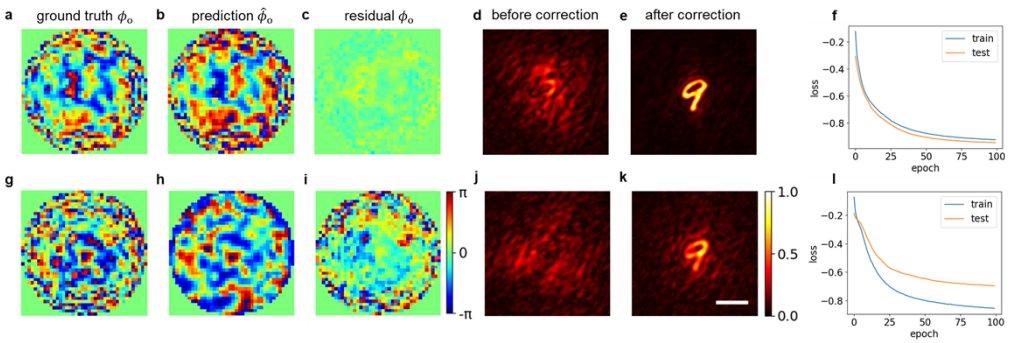

Fig. 2. Training results of the model trained using reflection matrices. (a-f) Results of the model trained on data with only output aberrations. (a) Ground truth aberration ϕ_o. (b) Predicted aberration $\hat{\phi}_o$. (c) Residual aberration $\phi_o - \hat{\phi}_o$. (d-e) Amplitude image at normal illumination before and after aberration correction. (f) Training and test loss throughout the model's training period. (g-l) Same as (a-f), but for the model trained on data with both output and input aberrations. Scale bar, 10 µm. Color bar, normalized amplitude by the maximum amplitude of aberration corrected image.

Training results from one of the test datasets are shown in Figs. 2(a)-(e). The ground truth aberration ϕ_o, predicted aberration $\hat{\phi}_o$, and residual aberration $\phi_o - \hat{\phi}_o$ are presented in Figs. 2(a)-(c), respectively. The residual aberration had an estimated Strehl ratio of 0.92, indicating a significant improvement from the initial aberration, which had a Strehl ratio of 0.04. The aberration correction was performed by dividing reflected wave by the predicted aberration $e^{i\hat{\phi}_o(\mathbf{k}_o)}$:

$$\tilde{E}_c(\mathbf{k}_o; \mathbf{k}_i) = e^{i(\phi_o(\mathbf{k}_o) - \hat{\phi}_o(\mathbf{k}_o))}\tilde{O}(\mathbf{k}_o - \mathbf{k}_i). \tag{5}$$

To illustrate the effect of aberration correction, we presented an amplitude image of the reflected wave under normal illumination $\mathbf{k}_i = (0, 0)$. After aberration correction, the image quality improved significantly, as shown in Fig. 2(e), compared to the uncorrected image in Fig. 2(d).

To evaluate the overall performance of our model, we measured the Strehl ratio of the residual aberration $\phi_o - \hat{\phi}_o$ for the test dataset. This evaluation yielded an average Strehl ratio of 0.85 ± 0.13 for the residual aberration. Compared to the initial ground truth aberration, which had an average Strehl ratio of 0.12 ± 0.10, this reflects a substantial reduction in aberration.

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In practical scenarios, the reflected waves experience both input and output aberrations. When the input aberration ϕ_i is present, it introduces a global phase shift to each reflected wave in reflection matrix, as illustrated in Fig. 1(b). As a result, inferring the common factor of output aberration $e^{i\phi_\mathrm{o}}$ becomes more complicated, as different incident angle produce different phase offset. Consequently, our previous model, trained on data containing only output aberration, is inadequate for this general case. To address this, we trained the model on data that incorporates both input and output aberrations, as described in Eq. (1).

Using the same U-Net architecture as in the previous model, we trained the network on computationally generated reflection matrices that included both input and output aberrations. The training dataset consisted of 1,000 samples, while the test dataset included 100 samples. The resulting loss curve is shown in Fig. 2(l). A notable discrepancy between the training and test data indicates imperfect training due to the presence of input aberrations.

As an illustrative example, the results from one of the test datasets are shown in Figs. 2(g)-(k). The ground truth aberration ϕ_o, predicted aberration $\hat{\phi}_\mathrm{o}$, and residual aberration $\phi_\mathrm{o} - \hat{\phi}_\mathrm{o}$ are presented in Figs. 2(g)-(i), respectively. The estimated Strehl ratio of the residual output aberration was 0.17, compared to the initial ground truth aberration, which had a Strehl ratio of 0.012. Although the Strehl ratio useful metric to assess wavefront quality, it becomes less reliable at such low-value regime. Therefore, for the initial aberration with a Strehl ratio of 0.012, we also report the root mean square (RMS) wavefront error, which was 1.70 rad. Figures 2(j) and (k) display the amplitude images before and after the aberration correction. On average, the Strehl ratio of the residual aberration $\phi_\mathrm{o} - \hat{\phi}_\mathrm{o}$ had a Strehl ratio of 0.44 ± 0.16, compared to an initial Strehl ratio of 0.11 ± 0.10. The achieved Strehl ratio is approximately half that of the case with only input aberration, confirming that this training approach is less effective when both input and output aberrations are present.

4. Iterative prediction and correction of both the input and output aberrations

Handling both input and output aberrations results in significantly poorer outcomes compared to correcting only output aberrations. In other words, the presence of input aberrations hampers the prediction of output aberrations. To overcome this, we propose an iterative correction method that alternates between correcting output and input aberrations (Fig. 3(a)). By leveraging the symmetry between the input and output of the reflection matrix, the same model designed for predicting output aberrations can also be used to predict input aberrations.

The process begins by predicting and correcting the output aberration, even if the correction is initially incomplete due to the influence of input aberrations. In the next step, the same model is applied to the transposed reflection matrix to predict and correct the input aberration. Although the first round of output aberration correction is only partial, it improves the accuracy of subsequent input aberration predictions. By iteratively refining both input and output aberrations, the model progressively achieves better performance as the opposing aberration is reduced, ultimately leading to more accurate predictions for both. The detailed mathematical description of this iterative correction procedure is described in the following.

The reflected waves associated with various incident wavevectors $\mathbf{k}_\mathrm{i}$ share a common output aberration $\phi_\mathrm{o}(\mathbf{k}_\mathrm{o})$. Similarly, the reflected waves corresponding to different reflected wavevectors $\mathbf{k}_\mathrm{o}$ share a common input aberration $\phi_\mathrm{i}(\mathbf{k}_\mathrm{i})$. Therefore, by changing the basis of the model input (i.e., the reflection matrix), from $\mathbf{k}_\mathrm{i}$ to $\mathbf{k}_\mathrm{o}$, the model can predict the input aberration $\phi_\mathrm{i}(\mathbf{k}_\mathrm{i})$, rather than the output aberration. This transformation is achieved by transposing the reflection matrix. Upon transposition, each column of the matrix becomes the complex field map $\tilde{E}(\mathbf{k}_\mathrm{i}; \mathbf{k}_\mathrm{o})$, which can be considered as the reflected wave measured at $\mathbf{k}_\mathrm{i}$ for a specific incident wavevectors $\mathbf{k}_\mathrm{o}$.

The detailed iterative correction process, illustrated in Fig. 3(a), is described as follows:

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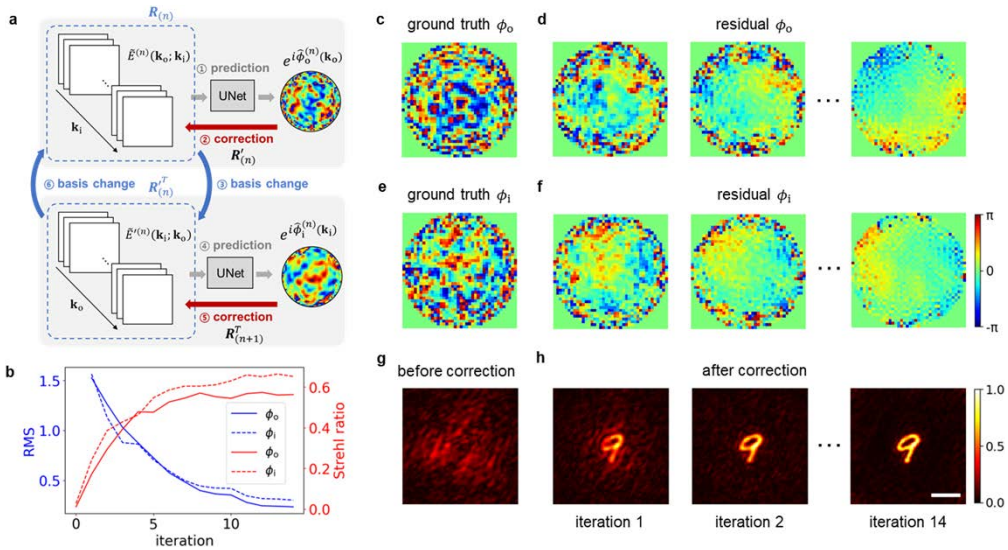

Fig. 3. Iterative correction using the trained model. (a) Diagram of iterative correction process. Initially, an output aberration, $\phi_o^{(n)}$ is predicted by input reflection matrix $\boldsymbol{R}_{(n)}$ into our model. The predicted output aberration is then used to correct the output aberration. The corrected matrix $\boldsymbol{R}'_{(n)}$ is then transposed to $\boldsymbol{R}'^{T}_{(n)}$ and forwarded to the model to predict the input aberration. This prediction is used to correct the input aberration. Subsequently, the matrix is again transposed to $\boldsymbol{R}_{(n+1)}$ for the next prediction of the output aberration. These steps are iteratively repeated until the stopping criterion is satisfied. (b) Strehl ratio of residual output (red solid) and input (red dashed) aberration at each iteration, with RMS of predicted output (blue solid) and input (blue dashed) aberrations. Iteration 0 represents the state prior to any correction, with the Strehl ratio at iteration 0 corresponding to that of the ground truth aberration. (c) Ground truth output aberration. (d) Residual output aberrations at iteration steps 1, 2, and 14. (e) Ground truth input aberration. (f) Residual input aberrations at iteration steps 1, 2, and 14. Color bar, phase in radians. (g) Amplitude image for normal illumination before aberration correction. (h) Aberration corrected images at iteration steps 1, 2, and 14. Scale bar, 10 µm. Color bar, normalized amplitude by the maximum amplitude at iteration steps 14.

1. Using the reflection matrix $\boldsymbol{R}_{(n)}$ at the nth stage of correction, predict the output aberration ϕ_o with the model:
$$e^{i\hat{\phi}_o^{(n)}(\mathbf{k}_o)} = UNet(\boldsymbol{R}_{(n)}). \tag{6}$$
2. Correct the output aberration based on this prediction. Correction is performed by multiplying complex conjugate of $\hat{\boldsymbol{P}}_o^{(n)}$, a diagonal matrix whose diagonal elements are $e^{i\hat{\phi}_o^{(n)}}$, by the reflection matrix:
$$\boldsymbol{R}'_{(n)} = \hat{\boldsymbol{P}}_o^{(n)*}\boldsymbol{R}(n). \tag{7}$$
3. Using the corrected reflection matrix $\boldsymbol{R}'_{(n)}$, predict the input aberration ϕ_i. As previously described, transpose the reflection matrix and input it into the same model:
$$e^{i\hat{\phi}_i^{(n)}(\mathbf{k}_i)} = UNet((\boldsymbol{R}'_{(n)})^T). \tag{8}$$
4. Correct the input aberration using this prediction. Correction is done by multiplying complex conjugate of $\hat{\boldsymbol{P}}_i^{(n)}$, a diagonal matrix with elements $e^{i\hat{\phi}_i^{(n)}}$, by the transposed

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reflection matrix:

$$\boldsymbol{R}^T_{(n+1)} = \hat{\boldsymbol{P}}^{(n)*}_{\mathrm{i}}(\boldsymbol{R}'_{(n)})^T. \tag{9}$$

After correction, the transposed reflection matrix $\boldsymbol{R}^T_{(n+1)}$ is again transposed to $\boldsymbol{R}_{(n+1)}$, in preparation for subsequent output aberration correction. By iteratively repeating this procedure, aberrations can be progressively reduced. However, the iterative correction algorithm does not inherently guarantee convergence, so appropriate stopping criteria are required. To determine when to stop, we monitored the RMS value of the predicted aberration map. A reduction in the RMS value indicates a decrease in residual aberration. Thus, iterations were terminated when the RMS values for both output and input aberrations no longer showed a declining trend (see Supplement 1 Section I for the details on the validity of the stopping criterion).

Figure 3 presents the results of the iterative correction applied to one of the test datasets. As shown in Fig. 3(b), the Strehl ratios of the residual aberrations in both the input and output domains progressively increased over 14 iterations, from 0.029 to 0.65 for the input, and from 0.012 (RMS wavefront error: 1.70 rad) to 0.56 for the output. The corresponding residual aberrations for the output and input are shown in Figs. 3(d) and 3(f), while the ground truth aberrations are displayed in Figs. 3(c) and 3(e). This iterative correction significantly enhanced image quality, as illustrated in Figs. 3(g)–(h).

We evaluated the iterative correction method on the test dataset and obtained average Strehl ratios of 0.68 ± 0.12 and 0.69 ± 0.16 for the residual input and output aberrations, respectively. These values reflect substantial improvements compared to the initial ground truth aberrations, which had Strehl ratios of 0.11 ± 0.10 and 0.10 ± 0.10. Given that the average Strehl ratio without iterative correction was only 0.44 ± 0.16 (equivalent to the first iteration), these results confirm that iterative correction significantly improves precision.

5. Training with complex object functions

In our previous models, the MNIST dataset was used as the target object for both training and testing. However, these models showed a significant decline in performance when applied to more complex target objects. To enhance the model's ability to handle a broader range of object functions, we trained it with a new dataset featuring increased complexity. Instead of the MNIST dataset, we used USAF targets as object functions and applied data augmentation techniques to increase training variability. These augmentations included random flipping, cropping, affine transformations, perspective transformations, and elastic transformations. Training was conducted using the same U-Net architecture with 4,000 training samples and 100 test samples.

To evaluate this model, we applied the iterative correction algorithm described in Section 5. Testing on the dataset showed that, after iterative correction, the average Strehl ratios for the input and output aberrations were 0.52 ± 0.12 and 0.52 ± 0.14, respectively. These values represent notable improvements over the initial ground truth aberrations, which had Strehl ratios of 0.11 ± 0.08 and 0.12 ± 0.10. Notably, the achieved Strehl ratio was lower than that obtained with the MNIST dataset, likely due to the increased complexity of the target objects.

Figures 4(a)–(d) illustrate an example of the model applied to data with a USAF target. After 15 iterations, the input and output aberrations were significantly reduced (Fig. 4(b)) compared to the ground truth (Fig. 4(a)), with Strehl ratios increasing from 0.086 and 0.072 to 0.73 and 0.75, respectively. This reduction in aberrations contributed to enhanced image quality, as shown in Fig. 4(c).The reconstructed image, further refined using CASS microscopy, is shown in Fig. 4(d).

Figure 4(e)–(l) demonstrates the model's generalization capability across diverse object functions beyond the USAF target. When applied to test datasets featuring a Siemens star target (Figs. 4(e)–(h)) and a custom-designed target (Figs. 4(i)–(l)), the input and output aberrations were substantially reduced. For the Siemens star, Strehl ratios improved from 0.10 and 0.24 to 0.70 and 0.81, respectively. For the custom target, they improved from 0.089 and 0.14 to 0.66

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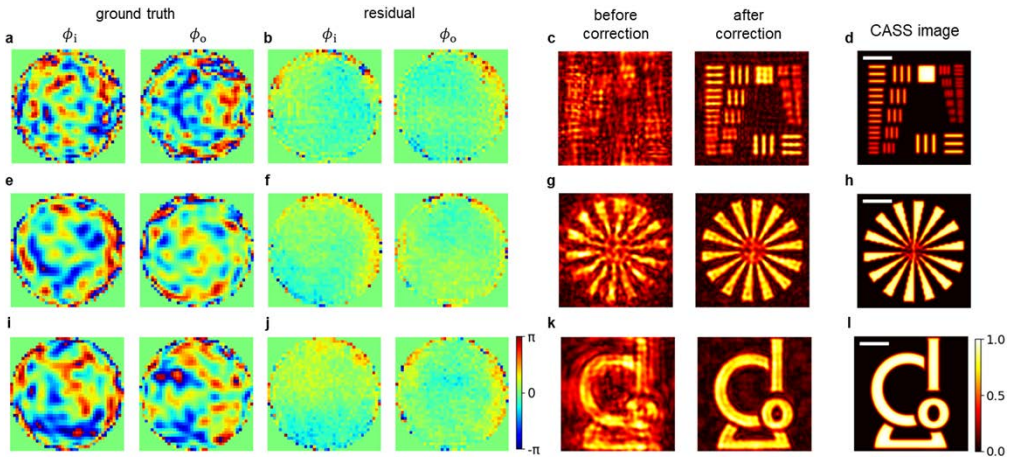

Fig. 4. Evaluation of model performance with different object functions. (a-d) The model trained with USAF target was tested on the data using the USAF target. (a) Ground truth input aberration ϕ_i and output aberration ϕ_o. (b) Residual input aberration $\phi_i - \hat{\phi}_i$ and output aberration $\phi_o - \hat{\phi}_o$ after 15 iterations. (c) Amplitude image for normal illumination before and after the iterative correction. (d) Reconstructed image using CASS microscopy after the iterative correction. (e-h) Same as (a-d), but the case when the same model was tested on the data using Siemens star target. (i-l) Same as (a-d), but the same model was tested on the data using custom-designed target. Scale bar, 10 µm. Color bar, normalized intensity.

and 0.72, respectively. These results highlight the model's robust performance in aberration correction for objects with spatial frequency spectrum characteristics distinct from the training data.

6. Covariance matrix iteration

Although training with the USAF target improved accuracy across various objects, the Strehl ratio only reached about 0.52, leaving room for further improvement. To address this, the covariance matrix, $\boldsymbol{RR}^\dagger$, was introduced as a model input, enhancing both prediction speed and accuracy. $\boldsymbol{RR}^\dagger$ matrix is constructed by multiplying reflection matrix $\boldsymbol{R}$ with its transpose conjugate [26]:

$$\boldsymbol{RR}^\dagger = (\boldsymbol{P}_o\boldsymbol{O}\boldsymbol{P}_i)(\boldsymbol{P}_o\boldsymbol{O}\boldsymbol{P}_i)^\dagger = \boldsymbol{P}_o\boldsymbol{O}\boldsymbol{P}_i\boldsymbol{P}_i^\dagger\boldsymbol{O}^\dagger\boldsymbol{P}_o^\dagger = \boldsymbol{P}_o\boldsymbol{O}\boldsymbol{O}^\dagger\boldsymbol{P}_o^\dagger = \boldsymbol{P}_o\boldsymbol{O}_I\boldsymbol{P}_o^\dagger. \tag{10}$$

Here $\boldsymbol{O}_I$ is a Toeplitz matrix with its elements determined by the Fourier transform of the intensity of object function, $|O(x, y)|^2$. The relation $\boldsymbol{P}_i\boldsymbol{P}_i^\dagger = \boldsymbol{I}$ holds, where $\boldsymbol{P}_i$ represents phase-only aberration. In the $\boldsymbol{RR}^\dagger$ matrix, the input aberration term is eliminated, allowing the input aberration prediction step in the iterative correction process to be skipped. This reduces the iteration time by half, as shown in Fig. 5(l).

We trained a new model using the $\boldsymbol{RR}^\dagger$ matrix with 4,000 training datasets and 100 test datasets, utilizing the same augmented USAF targets and U-Net architecture as in the previous model. In this configuration, the $\boldsymbol{RR}^\dagger$ matrix replaced the reflection matrix $\boldsymbol{R}$ as the model input. For evaluation, we applied the iterative correction algorithm to the test dataset. The algorithm for the $\boldsymbol{RR}^\dagger$ matrix is similar to that for the reflection matrix, with the key difference being the omission of input aberration prediction and correction, as the $\boldsymbol{RR}^\dagger$ matrix contains only the output aberration term. Results showed that after iterative correction, the average Strehl ratio for residual output aberrations improved to 0.72±0.18 from 0.52±0.14 of the model trained with the $\boldsymbol{R}$ matrix.

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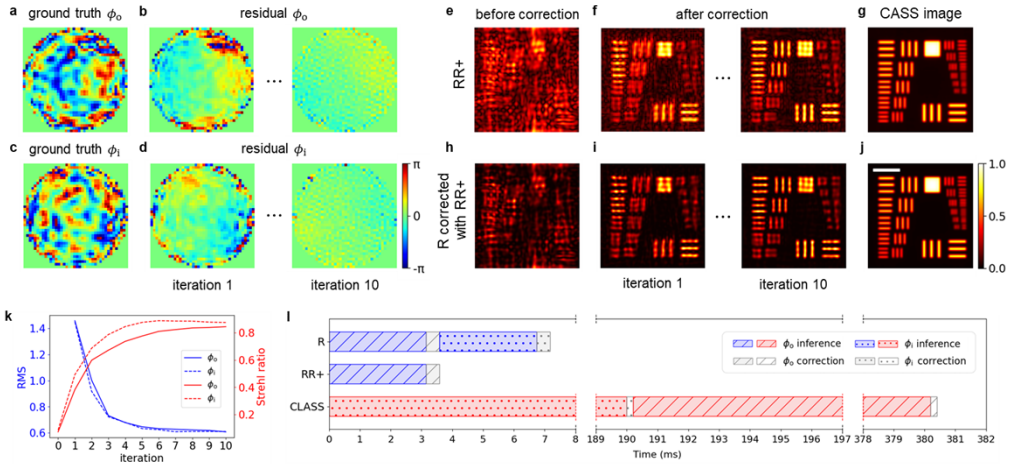

Fig. 5. $\boldsymbol{RR}^\dagger$-trained models and time evaluation. (a) Ground truth output aberration ϕ_o. (b) Residual output aberrations $\phi_o - \hat{\phi}_o$ at iteration steps 1 and 10. (c) Ground truth input aberration ϕ_i. (d) Residual input aberrations $\phi_i - \hat{\phi}_i$ at iteration steps 1 and 10. (e-g) Iterative correction results of the $\boldsymbol{RR}^\dagger$-only method. (e) Intensity image for normal illumination before aberration correction. (f) Aberration corrected images at iteration steps 1 and 10. (g) Reconstructed image using CASS microscopy after the iterative correction. (h-j) Iterative correction results of the combined method, which corrects $\boldsymbol{R}$ using aberrations identified with $\boldsymbol{RR}^\dagger$ and $\boldsymbol{R}^\dagger\boldsymbol{R}$. (h) Intensity image for normal illumination before aberration correction. (i) Aberration corrected images at iteration steps 1 and 10. Color bar, normalized intensity by the maximum intensity at iteration 10. (j) Reconstructed image using CASS microscopy after the iterative correction. Scale bar, 10 µm. Color bar, normalized intensity by its maximum intensity. (k) Strehl ratio of residual output (red solid) and input (red dashed) aberration at each iteration, with the RMS of the predicted output (blue solid) and input (blue dashed) aberration. Iteration 0 represents the state prior to any correction, with the Strehl ratio at iteration 0 corresponding to that of the ground truth aberration. (l) Time evaluation of the iterative correction processes. For the $\boldsymbol{R}$-method, the process for each iteration includes: output aberration inference (blue dashed), output aberration correction (gray dashed), input aberration inference (blue dotted), and input aberration correction (gray dotted), totaling 7.2 ms. For the $\boldsymbol{R}^\dagger\boldsymbol{R}$-methods, the process consists of: output aberration inference (blue dashed), output aberration correction (gray dashed) totaling 3.6 ms. The CLASS algorithm includes: output aberration inference (red dashed), output aberration correction (white dashed), input aberration inference (red dotted), and input aberration correction (white dotted), totaling 380.4 ms.

As an illustrative example, the model was tested using the $\boldsymbol{RR}^\dagger$ matrix derived from the same reflection matrix used in Figs. 4(a)-(d). After 10 iterations, the ground truth output aberrations (Fig. 5(a)) were reduced (Fig. 5(b)), with the Strehl ratio improving from 0.072 to 0.84, surpassing the performance of the model trained with the $\boldsymbol{R}$ matrix, which achieved a Strehl ratio of 0.75. The Strehl ratio progression over the iterations is shown as a solid red line in Fig. 5(k). Additionally, Figs. 5(e) and (f) show single-shot intensity images obtained from the $\boldsymbol{RR}^\dagger$ matrix before and after aberration correction, demonstrating improvement over iterations.

Although the $\boldsymbol{RR}^\dagger$ matrix offers improved prediction accuracy, its configuration inherently imposes a limitation on resolution. This limitation arises because the intensity of the object function is encoded within a matrix with only half the resolution of the original during the construction of the $\boldsymbol{RR}^\dagger$ matrix (see Supplement 1 Section II for details). To overcome this limitation, we propose a strategy to predict aberrations using model trained with the $\boldsymbol{RR}^\dagger$ for

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better accuracy and subsequently correct these aberrations to the original reflection matrix for maintaining optimal resolution. In the original reflection matrix, aberrations are present in both the input and output channels, necessitating separate predictions for each. We can predict the output aberration using the $\boldsymbol{RR}^\dagger$ matrix. For predicting the input aberration, we constructed the $\boldsymbol{R}^\dagger\boldsymbol{R}$ matrix, as follows:

$$\boldsymbol{R}^\dagger\boldsymbol{R} = (\boldsymbol{P}_\mathrm{o}\boldsymbol{O}\boldsymbol{P}_\mathrm{i})^\dagger(\boldsymbol{P}_\mathrm{o}\boldsymbol{O}\boldsymbol{P}_\mathrm{i}) = \boldsymbol{P}_\mathrm{i}{}^\dagger\boldsymbol{O}^\dagger\boldsymbol{P}_\mathrm{o}^\dagger\boldsymbol{P}_\mathrm{o}\boldsymbol{O}\boldsymbol{P}_\mathrm{i} = \boldsymbol{P}_\mathrm{i}{}^\dagger\boldsymbol{O}\boldsymbol{O}^\dagger\boldsymbol{P}_\mathrm{i} = \boldsymbol{P}_\mathrm{i}\boldsymbol{O}_I\boldsymbol{P}_\mathrm{i}^\dagger. \tag{11}$$

By using this $\boldsymbol{R}^\dagger\boldsymbol{R}$ matrix as the input to the same model trained with the $\boldsymbol{RR}^\dagger$ matrix, it becomes possible to predict the input aberration instead of the output aberration.

Using the $\boldsymbol{R}^\dagger\boldsymbol{R}$ matrix, the ground truth input aberrations (Fig. 5(c)) from the same reflection matrix used in Figs. 4(a)-(d) were reduced (Fig. 5(d)), with the Strehl ratio improving from 0.086 to 0.88, again surpassing the performance of the model trained with the $\boldsymbol{R}$ matrix, which achieved a Strehl ratio of 0.73. The Strehl ratio for input aberrations over the iterations is shown as a dotted red line in Fig. 5(k). Figure 5(h) shows the intensity image under normal illumination before correction, and Fig. 5(i) shows the image after correcting both input and output aberrations found with $\boldsymbol{R}^\dagger\boldsymbol{R}$ and $\boldsymbol{RR}^\dagger$, respectively. Figures 5(g) and (j) show the reconstructed CASS images of the $\boldsymbol{RR}^\dagger$-only method and the combined method, which corrects $\boldsymbol{R}$ using aberrations identified with $\boldsymbol{RR}^\dagger$ and $\boldsymbol{R}^\dagger\boldsymbol{R}$. These results confirm that the second method provides enhanced image quality with higher resolution.

7. Time evaluation

Our deep-learning-based iterative correction algorithm offers a significant advantage in computation speed. Specifically, our approach is 100 times faster than the traditional correlation-based CLASS algorithm. Both the deep-learning-based and CLASS algorithms share a similar structural framework. They operate by first predicting and correcting the aberration on one side (either input or output), then predicting and correcting the aberration on the other side. This process repeats until convergence.

To evaluate computational speed, we compared the inference and correction times per iteration for both methods, as shown in Fig. 5(l). A single inference step using our model, corresponding to the model's inference time, takes approximately 3.2 ms, whereas the CLASS algorithm requires 190 ms for the equivalent correlation calculation. While our model includes an additional step to convert between real and complex values—resulting in a correction time of 0.4 ms, compared to 0.2 ms for CLASS—the correction time is negligible relative to inference time. Consequently, the total time per iteration is 7.2 ms for our approach, compared to 380.4 ms for CLASS. Moreover, when using our model with the $\boldsymbol{RR}^\dagger$ matrix, we can bypass input aberration correction, further reducing the iteration time to 3.6 ms, achieving a 100-fold speed improvement over CLASS. Additionally, in the combined method that delivers the best performance, iterations for the $\boldsymbol{RR}^\dagger$ and $\boldsymbol{R}^\dagger\boldsymbol{R}$ matrices are independent, allowing for parallel execution on GPUs. As a result, the computational time remains identical to the $\boldsymbol{RR}^\dagger$-only method, though memory requirement is doubled.

8. Validation with experimental data

We applied our deep-learning models with the iterative correction algorithm to experimental data. The reflection matrix was obtained using the setup described in [27]. To introduce aberrations, we prepared a sample consisting of a USAF target placed under a curved plastic layer. The aberrating layer was prepared by manually crumpling a 270 μm-thick plastic lid and positioning it ~2 mm above the target. Figure 6 compares the results of our method in various configurations with those of the CLASS algorithm. Since ground truth data is unavailable for the experimental case, we use CLASS results (Figs. 6(c) and 6(d)) as a reference for comparison, examining the

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predicted aberration and the corresponding reconstructed images. Additionally, the experimental data is affected by both aberrations and various noise sources, including multiple scattering and system noise. To mitigate these effects, we displayed CASS image rather than a single image taken at normal illumination. For reference, Fig. 6(b) presents the reconstructed CASS image before aberration correction.

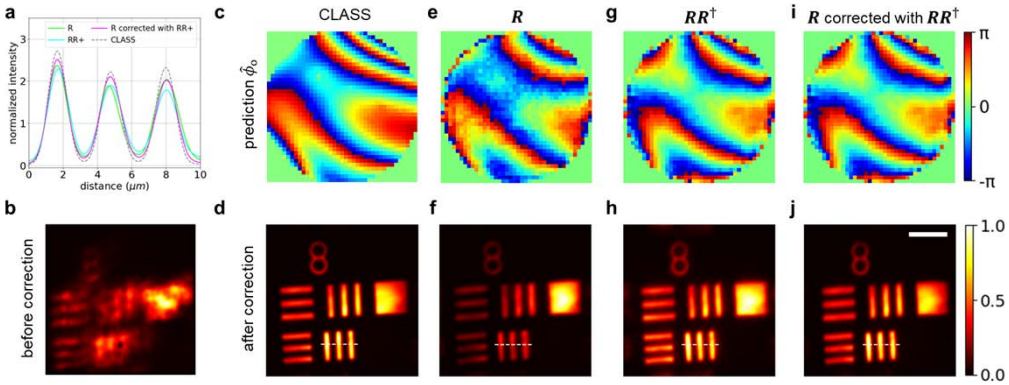

Fig. 6. Experimental demonstration. (a) Intensity profile along the dashed line in (b)-(j). (b) CASS image before aberration correction. (c-j) The predicted output aberrations and corresponding CASS images obtained using the CLASS algorithm ((c) and (d)), the model trained with $\boldsymbol{R}$ ((e) and (f)), the $\boldsymbol{RR}^\dagger$-only method ((g) and (h)), and the combined method ((i) and (j)). Scale bar, 10 μm. Color bar, normalized intensity relative to its maximum intensity.

Applying the model trained with $\boldsymbol{R}$, the results (Figs. 6(e) and (f)) are similar to those of CLASS but exhibit slight degradation on the left side of the reconstructed image. Figure 6(g) and (h) present the results using the $\boldsymbol{RR}^\dagger$-only method, which directly corrects $\boldsymbol{RR}^\dagger$ based on aberrations identified within $\boldsymbol{RR}^\dagger$. This method improves reconstruction quality on the left side, but compromises resolution compared to CLASS as expected. The results improved further by applying the combined method, which corrects $\boldsymbol{R}$ using aberrations identified with $\boldsymbol{RR}^\dagger$ and $\boldsymbol{R}^\dagger\boldsymbol{R}$, as illustrated in Figs. 6(i) and (j). This approach achieved performance comparable to the CLASS algorithm. For a detailed comparison, Fig. 6(a) presents the line profiles of the results obtained using these methods along the white lines shown in Figs. 6(d)–(j), highlighting their respective performance.

9. Discussion

This study introduces a deep learning-based model for aberration prediction and correction in reflection matrix microscopy by using a U-Net architecture that captures the consistent aberration factors across different incident waves. The model effectively predicts aberrations directly from the reflection matrix, and through an iterative correction strategy, it alternates between correcting output and input aberrations. This iterative approach progressively refines the model's predictions, leading to enhanced accuracy. Notably, by leveraging the covariance matrix method, the computation time is significantly reduced, enhancing both accuracy and iteration efficiency. As a result, our approach achieves a 100-fold speed improvement compared to traditional CLASS algorithms. This substantial acceleration is crucial for real-time aberration correction and dynamic adjustments in systems such as spatial light modulators.

Our method demonstrated excellent performance not only on synthetic datasets but also on experimental data, validating its robustness and applicability in real-world settings. This deep learning-based framework offers the potential for more efficient and precise aberration

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correction, which is vital for high-resolution imaging in dynamic environments. Moreover, by using simulation-generated datasets, the model generalizes well across different types of objects, demonstrating the effectiveness of the approach in various practical scenarios.

Beyond the scope of reflection matrix-based aberration correction, this method holds significant promise for extending to other imaging modalities. For instance, its potential application to multiple scattering tracing (MST) [28] algorithms, second-harmonic microscopy [29], and fluorescence imaging (via the incoherent response matrix, IRM) [30] points to the model's flexibility. These applications could expand the reach of this deep-learning approach in both coherent and incoherent imaging systems, marking a significant step forward in biomedical research and diagnostics.

While the current model works well in environments where aberrations arise primarily due to incident-angle-based phase retardations with minimal scattering, challenges remain in more complex settings. In imaging environments where severe aberrations and scattering coexist, the current deep-learning model faces limitations, while CLASS algorithms continue to perform effectively. In particular, when the noise-to-signal intensity ratio exceeds 4, the model's performance begins to degrade significantly, especially in recovering high-frequency aberration components (see Supplement 1 Section III for a detailed analysis of noise robustness). Thus, future research should focus on advancing deep learning models capable of addressing complex aberrations in highly scattering media, further enhancing the model's robustness.

In this work, we adopted a U-Net-based architecture due to its proven performance in image-to-image prediction tasks. While this architecture provided stable and accurate results in our experiments, we acknowledge that more advanced architectures—such as attention-based networks or transformer-enhanced models—may offer further performance gains. Exploring these alternatives remains an important direction for future research.

To conclude, this study represents a crucial step toward integrating deep learning with reflection matrix microscopy for real-time aberration correction. By striking a balance between computational efficiency and imaging precision, the proposed method opens doors for label-free imaging in biomedical applications. Future efforts will aim to refine the model further, tackling more complex optical phenomena, and thus enhancing the broader impact of this approach across a variety of imaging fields.

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Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

Supplemental document. See Supplement 1 for supporting content.

References

1. M. J. Booth, "Adaptive optical microscopy: the ongoing quest for a perfect image," Light:Sci. Appl. **3**(4), e165 (2014).
2. N. Ji, D. E. Milkie, and E. Betzig, "Adaptive optics via pupil segmentation for high-resolution imaging in biological tissues," Nat. Methods **7**(2), 141–147 (2010).
3. P. N. Marsh, D. Burns, and J. M. Girkin, "Practical implementation of adaptive optics in multiphoton microscopy," Opt. Express **11**(10), 1123–1130 (2003).
4. J. H. Park, W. Sun, and M. Cui, "High-resolution in vivo imaging of mouse brain through the intact skull," Proc. Natl. Acad. Sci. U.S.A. **112**(30), 9236–9241 (2015).
5. C. Wang, R. Liu, D. E. Milkie, *et al.*, "Multiplexed aberration measurement for deep tissue imaging in vivo," Nat. Methods **11**(10), 1037–1040 (2014).
6. M. J. Booth, M. A. A. Neil, R. Juškaitis, *et al.*, "Adaptive aberration correction in a confocal microscope," Proc. Natl. Acad. Sci. U.S.A. **99**(9), 5788–5792 (2002).

Optics EXPRESS

7. R. Fiolka, K. Si, and M. Cui, "Complex wavefront corrections for deep tissue focusing using low coherence backscattered light," Opt. Express **20**(15), 16532–16543 (2012).
8. S. Tuohy and A. G H. Podoleanu, "Depth-resolved wavefront aberrations using a coherence-gated Shack-Hartmann wavefront sensor," Opt. Express **18**(4), 3458–3476 (2010).
9. M. Rueckel, J. A. Mack-Bucher, and W. Denk, "Adaptive wavefront correction in two-photon microscopy using coherence-gated wavefront sensing," Proc. Natl. Acad. Sci. U.S.A. **103**(46), 17137–17142 (2006).
10. J. Wang, J.-F. Léger, J. Binding, *et al.*, "Measuring aberrations in the rat brain by coherence-gated wavefront sensing using a Linnik interferometer," Biomed. Opt. Express **3**(10), 2510–2525 (2012).
11. X. Tao, A. Norton, M. Kissel, *et al.*, "Adaptive optical two-photon microscopy using autofluorescent guide stars," Opt. Lett. **38**(23), 5075–5078 (2013).
12. S. Kang, P. Kang, S. Jeong, *et al.*, "High-resolution adaptive optical imaging within thick scattering media using closed-loop accumulation of single scattering," Nat. Commun. **8**(1), 2157 (2017).
13. P. Zhang, S. Liu, A. Chaurasia, *et al.*, "Analyzing complex single-molecule emission patterns with deep learning," Nat. Methods **15**(11), 913–916 (2018).
14. Y. Jin, Y. Zhang, L. Hu, *et al.*, "Machine learning guided rapid focusing with sensor-less aberration corrections," Opt. Expr. **26**(23), 30162–30171 (2018).
15. L. Möckl, P. N. Petrov, and W. E. Moerner, "Accurate phase retrieval of complex 3D point spread functions with deep residual neural networks," Appl. Phys. Lett. **115**(25), 251106 (2019).
16. B. P. Cumming and M. Gu, "Direct determination of aberration functions in microscopy by an artificial neural network," Opt. Express **28**(10), 14511 (2020).
17. I. Vishniakou and J. D. Seelig, "Wavefront correction for adaptive optics with reflected light and deep neural networks," Opt. Express **28**(10), 15459 (2020).
18. P. Khorin, A. P. Dzyuba, P. G. Serafimovich, *et al.*, "Neural networks application to determine the types and magnitude of aberrations from the pattern of the point spread function out of the focal plane," J. Phys.: Conf. Ser. **2086**(1), 012148 (2021).
19. D. Saha, D. Saha, U. Schmidt, *et al.*, "Practical sensorless aberration estimation for 3D microscopy with deep learning," Opt. Express **28**(20), 29044–29053 (2020).
20. E. Durech, W. Newberry, J. Franke, *et al.*, "Wavefront sensor-less adaptive optics using deep reinforcement learning," Biomed. Opt. Express **12**(9), 5423–5438 (2021).
21. E. Bostan, R. Heckel, M. Chen, *et al.*, "Deep phase decoder: self-calibrating phase microscopy with an untrained deep neural network," Optica **7**(6), 559–562 (2020).
22. Q. Xin, Q. Xin, G. Ju, *et al.*, "Object-independent image-based wavefront sensing approach using phase diversity images and deep learning," Opt. Express **27**(18), 26102–26119 (2019).
23. Q. Hu, M. Hailstone, J. Wang, *et al.*, "Universal adaptive optics for microscopy through embedded neural network control," Light: Sci. Appl. **12**(1), 270 (2023).
24. S. Kang, S. Jeong, W. Choi, *et al.*, "Imaging deep within a scattering medium using collective accumulation of single-scattered waves," Nat. Photonics **9**(4), 253–258 (2015).
25. O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional Networks for Biomedical Image Segmentation," Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics **9351**, 234–241 (2015).
26. H. Lee, S. Yoon, P. Loohuis, *et al.*, "High-throughput volumetric adaptive optical imaging using compressed time-reversal matrix," Light: Sci. Appl. **11**(1), 16 (2022).
27. Y. R. Lee, D. Y. Kim, Y. Jo, *et al.*, "Exploiting volumetric wave correlation for enhanced depth imaging in scattering medium," Nat. Commun. **14**(1), 1878 (2023).
28. S. Kang, Y. Kwon, H. Lee, *et al.*, "Tracing multiple scattering trajectories for deep optical imaging in scattering media," Nat. Commun. **14**(1), 6871 (2023).
29. S. Yoon, H. Lee, J. H. Hong, *et al.*, "Laser scanning reflection-matrix microscopy for aberration-free imaging through intact mouse skull," Nat. Commun. **11**(1), 5721 (2020).
30. S. Lim, S. Kang, J.-H. Hong, *et al.*, "Multiphoton super-resolution imaging via virtual structured illumination," arXiv (2024).

163.152.3.135 2025/06/18 02:58:15